



西子奥的斯电梯有限公司

AMCB2

No: XAA610BB_GL_EN

Date: 04/07/2006

Page: 1/24

Guide Lines



AMCB2

Guide lines

Xizi Otis Elevator Co., Ltd

No: XAA610BB_GL_EN



西子奥的斯电梯有限公司

AMCB2

Guide Lines

No: XAA610BB_GL_EN

Date: 04/07/2006

Page: 2/24

PC Record

Edit	Xia yongjie	Check	Liu Zubin	Auth.	Qian Xinhui	Appr.	Zhang Hongbing
Dept.	PDC	Dept.	PDC	Dept.	EC	Dept.	PDC
Date	2006-7-5	Date	2006-7-5	Date	2006-7-5	Date	2006-7-5

AUTH. PC-LS0010-2006

Component Version

Name	Hardware	Software
AMCB2	1.0	3.5



AMCB2

No: XAA610BB_GL_EN

Date: 04/07/2006

Page: 3/24

Guide Lines

西子奥的斯电梯有限公司

Modification Record:

No	PC	Content	Date	Signature
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

This work and the information it contains are the property of Xizi Otis Elevator Company ("XOEC"). It is delivered to others on the express condition that it will be used only for, or on behalf of, XOEC; that neither it nor the information it contains will be reported or disclosed, in whole or in part, without the prior written consent of XOEC, and that on demand it and any copies will be promptly returned to XOEC.

Contents

1	General.....	3
2	Top view of AMCB2:.....	3
2.1	LCD display	3
2.1.1	Monitor.....	3
2.2	Operate Keys	3
2.3	Pin Assignments.....	3
3	Run sequence.....	3
3.1	Inspection Run.....	3
3.2	Normal run.....	3
3.3	Correct run.....	3
3.4	Learning run.....	3
3.5	NTSD protection run.....	3
4	Parameters list introduce.....	3
4.1	System parameter F0.....	3
4.2	Parameters of run curveF1~F6.....	3
4.3	Related parameters about floor, speed and so on.....	3
4.4	The parameters about adjustment of leveling position.....	3
4.5	The parameters of adjusting load weighting device.....	3
4.6	Parameters about adjusting the brake delay:.....	3
5	Other function parameters.....	3
6	Troubleshooting.....	3

	AMCB2	No: XAA610BB_GL_EN
	Guide Lines	Date: 04/07/2006 Page: 5/24

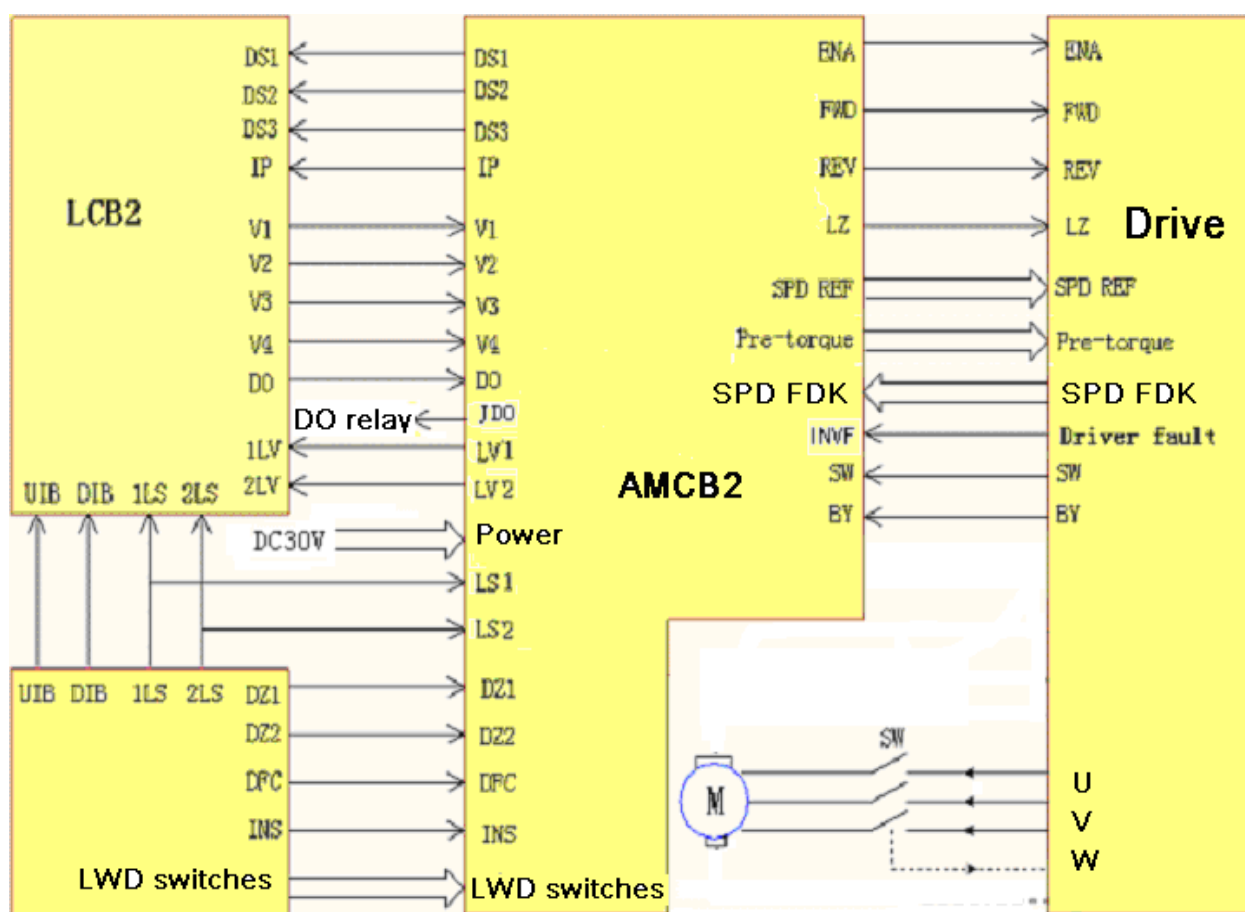
西子奥的斯电梯有限公司

General:

AMCB2 board is used to the OH-CON4421/3 control system of XOEC, and those systems used in XO-STAR and OH5000 elevators.

The board is a bridge between the OTIS LCBII control board and general inverter. It receives the V codes form LCBII, and then switches the V codes to run command and speed curve to drive. It also monitors the inverter status, and sends DS (drive statue) codes to LCBII

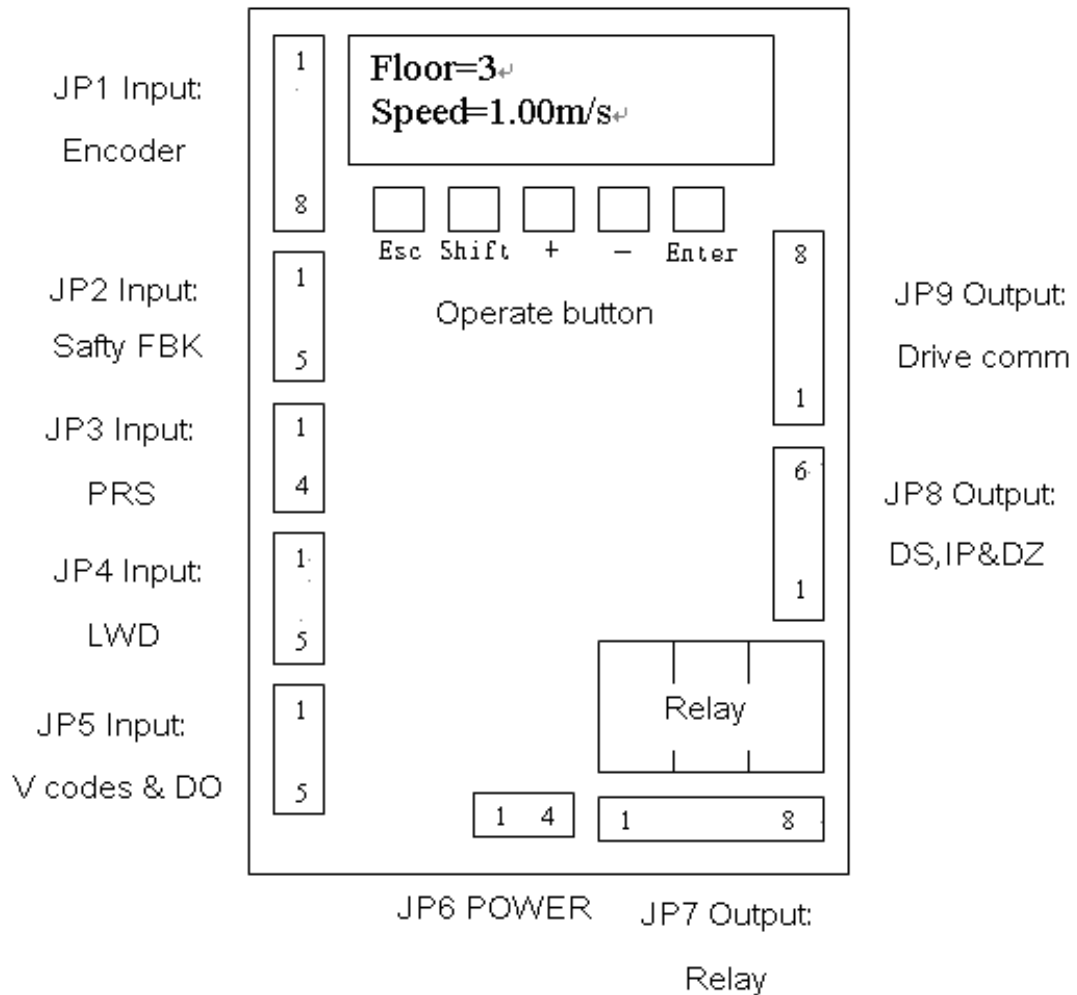
System is shown below:



Guide Lines

西子奥的斯电梯有限公司

Top view of AMCB2:



Guide Lines

西子奥的斯电梯有限公司

1.1 LCD display

the display of AMCB2 has three interfaces: monitor, menu and trouble (only display when the fault happens).

1.1.1 Monitor

the Monitor interface shows the car current position and speed of car.

1.1.2 Menu

The Menu interface shows two menus: F menu, P menu.

The F menu is function parameters, the basic parameter of elevator run like speed curve, contract speed and inspection speed and so on are in this menu;

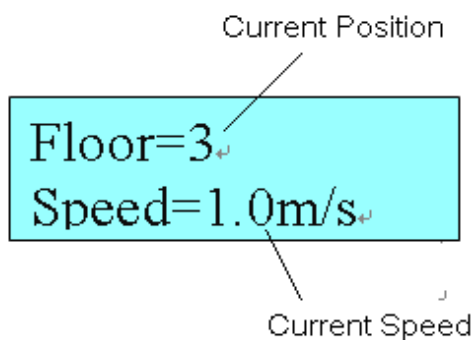
The P menu is position parameters, all hoistway learning parameters are saved in it. For example: all landing position, 1LS/2LS position, the length of vane and distance of two top of car sensors.

1.1.3 Trouble

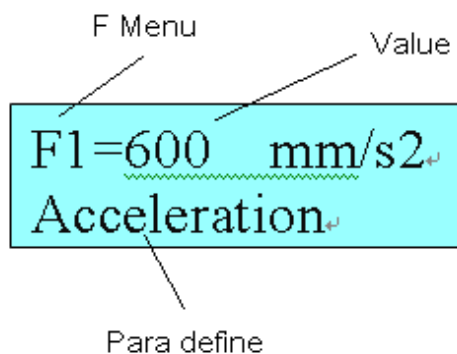
The trouble menu shows the entire fault, it will be automatically display when the fault happened. The faults include encoder fault, inverter fault and contactor fault etc.

The below table are three interface:

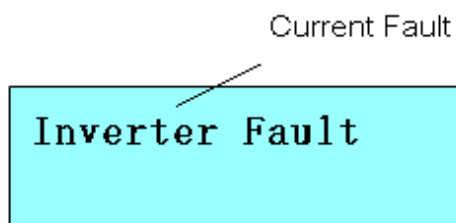
Monitor:



Menu:



Trouble:



1.2 Operate Keys

There are 5 operate keys under the displayer, it is used:

Modify the parameters of F menu, switch P/F menu each other

Name	Function
Esc	Used to go up one level in menu navigation; Used to escape out of parameters edit or selection with no change.
Shift	Scroll digits in parameters edit mode; P menu and F menu exchange
+	Used to scroll up menu items in menu navigation; Added the value in Parameters edit mode
-	Used to scroll down menu items in menu navigation; Reduced the value in Parameters edit mode
Enter	Used to enter the selected parameter or parameter value after editing.

西子奥的斯电梯有限公司

1.3 Pin Assignments

The AMCB2 is applicable for two Drive Types; when it uses different drives, some terminals assignment will be different.

Plug JP1	1	2	3	4	5	6	7	8
Signal	A+	A-	B+	B-	GND	A	B	0V
type	In	In	In	In	In	In	In	In
Volt. garde	5Vdc					24Vdc		
Function	Encoder signal input(line driver input)					Encoder signal (complementary)		

- JP1-1~5: used to input line driver encoder signal; e.g.: EXP-E of SIEI drive;
- JP1-6~8: use to input complementary encoder signal; e.g.: PG-B2 of YASKAWA drive.

Plug JP2	1	2	3	4	5
Signal	DFC	INS	INVF	SW	BY
type	In	In	In	In	In
Volt. garde	19.2~40Vdc				
Function	Door full close	Inspection run	Drive fault	Enable contact	Enable brake

Plug JP3	1	2	3	4
Signal	1LV	2LV	1LS	2LS
Type	In	In	In	In
Volt. garde	19.2~40Vdc			
Function	Up sensor	Down sensor	1 limit SW	2 limit SW

Plug JP4		1	2	3	4	5
Signal	DRIVE=0	LWS	LNS	50L	25L	ANS
	DRIVE=2	SW FBK	LNS	50L	30L	BY FBK
Type		In	In	In	In	In
Volt. garde		19.2~40Vdc				
Function	DRIVE=0	100%	80%	50%	25%	10%
	DRIVE=2	SW FBK	80%	50%	30%	BY FBK

Plug JP5		1	2	3	4	5
Signal		V1	V2	V3	V4	DO
Type		IN	IN	IN	IN	IN
Volt. garde		24Vdc				
Function		Velocity codes				Enable DO

➤ The input of velocity codes are coded as described below:

Mode	Function	V4	V3	V2	V1
WT	wait	0	0	0	0
ST	stop	1	1	1	1
IR-U	Inspection run up	0	1	0	0
IR-D	Inspection run down	0	1	0	1
FR-U	Normal run up	0	1	1	0
FR-D	Normal run down	0	1	1	1
RL-U	Relevel run up	1	0	1	0

Guide Lines

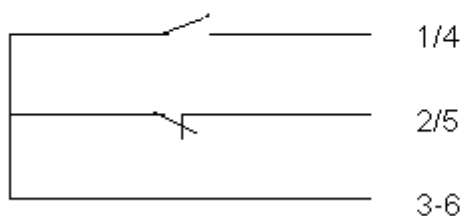
西子奥的斯电梯有限公司

RL-D	Relevel run down	1	0	1	1
SD	Slow down	1	1	1	0
-	(Not allowed)				

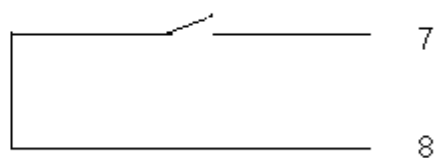
Plug JP6	1	2	3	4
Signal	30VDC	30VDC	0V	0V
Type	In	In	In	In
Volt. garde	24~40Vdc			
Function	Power supply			

Plug JP7	1	2	3	4	5	6	7	8
Signal	DO Relay			BY Relay			SW Relay	
Type	Out			Out			Out	
Volt. garde	24Vdc 5A\110/220Vac 5a							
Function	Door open relay			Brake relay			contactor	

DO/BY Relay



SW Contactor



Plug JP8	1	2	3	4	5	6
Signal	DS1	DS2	DS3	IP	1LV	2LV

Guide Lines

西子奥的斯电梯有限公司

Type	Out	Out	Out	Out	Out	Out
Volt grade	24Vdc 100mA					
Function	Drive status codes		IP	Up sensor	Down sensor	

➤ DS(Drive status)

DS3	DS2	DS1	Remark
0	0	0	Drive fault
0	0	1	Speed > 0.3m/s,
0	1	0	Normal run
1	0	0	Learn run
1	1	0	speed<0.3m/s
1	1	1	stop
other			-

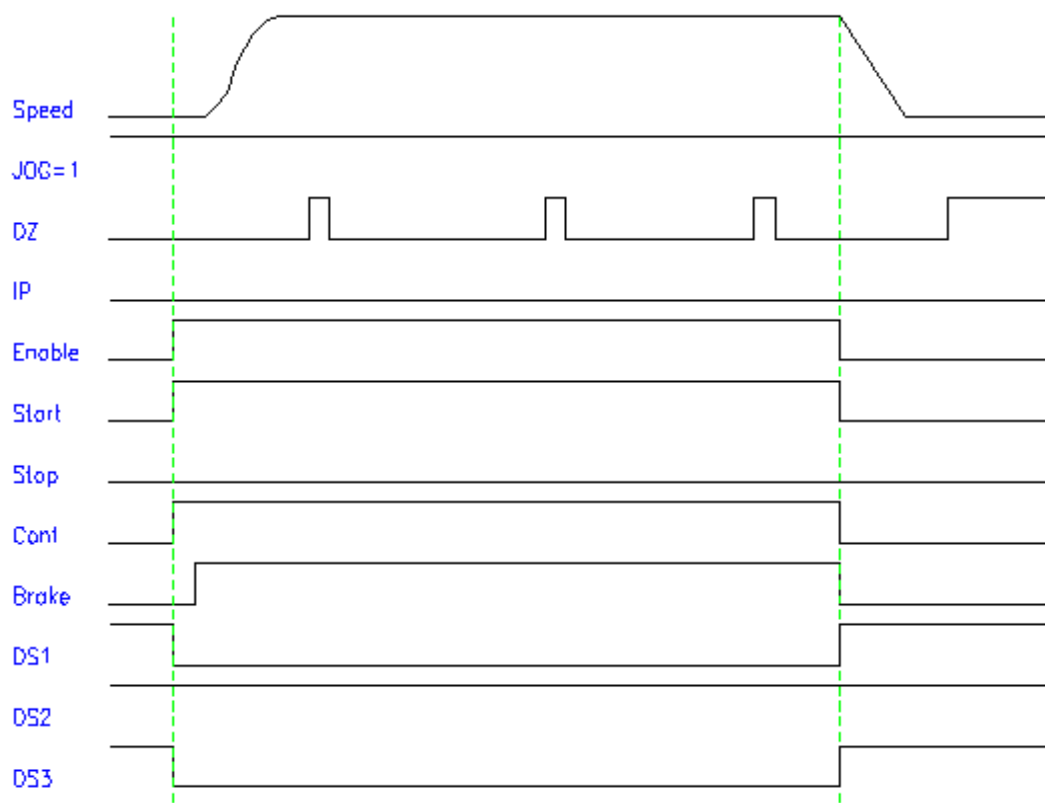
Plug JP9	1	2	3	4	5	6	7	8
Signal	ENA	FWD	REV	LZ	COM	VREF	VTOQ	0V
Type	Out	Out	Out	Out	Out	Out	Out	Out
Volt grade	24Vdc 100mA					0~10Vdc 20mA		
Function	Enable	Forward	reverse	Landing zone	Com of output	Velocity Analog output	LWD analog output	Com

西子奥的斯电梯有限公司

2 Run sequence

2.1 Inspection Run

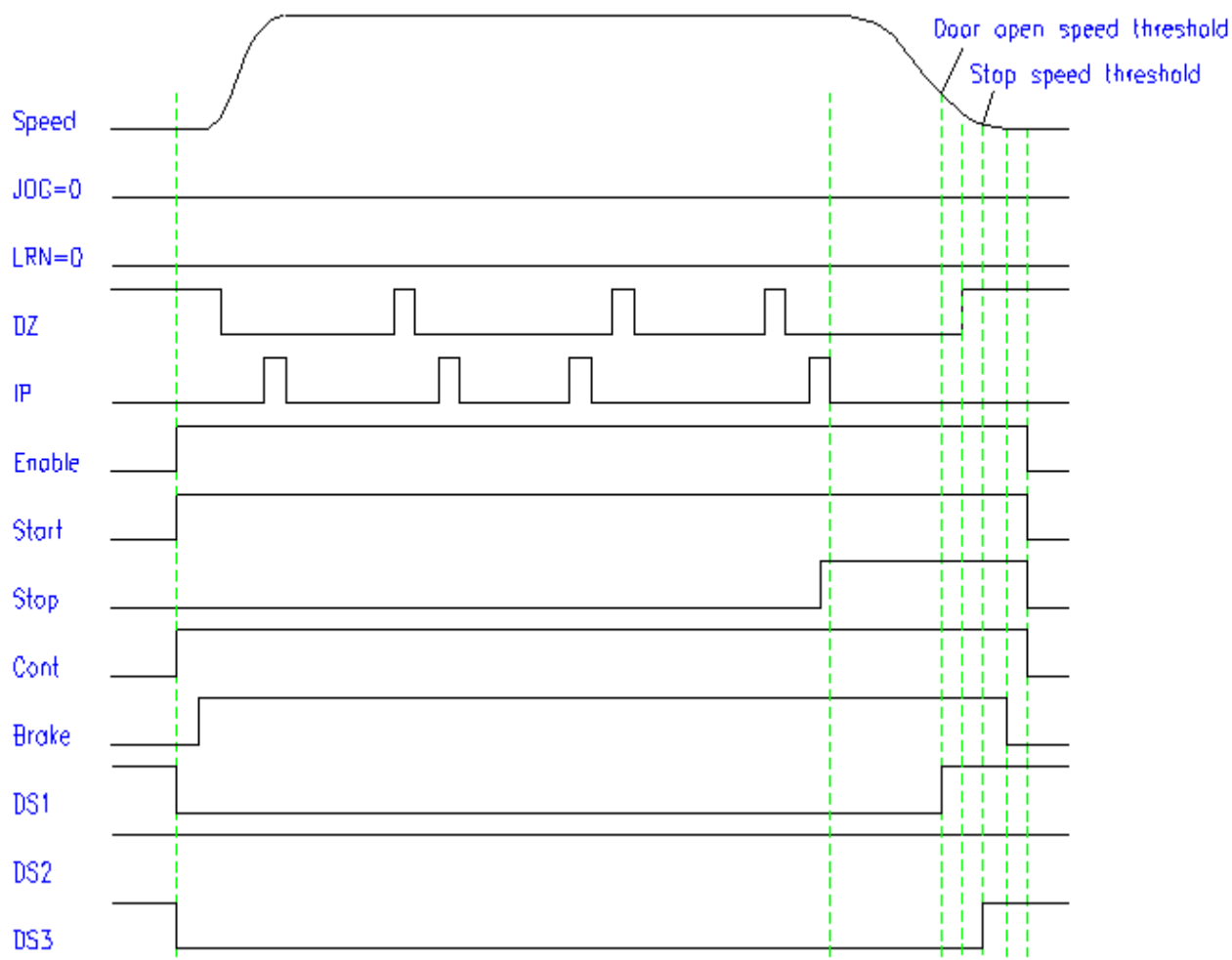
Inspection Run



- If the JOG (inspection) is activated, the AMCB2 is in the inspection mode.
- AMCB2 output the Enable and Start (FWD/REV) signals according V code; and after the BY signal is activated, the AMCB2 start to output the speed curve.
- in inspection run mode, the IP signal will never be activated
- when the lift start to inspection run, the AMCB2 change the DS codes form 111 to 010 to LCBII; when the lift stop to run, the Enable and Start signals is off and DS codes change to 111 to LCBII.
- When the lift is in inspection run mode, and If Enable or Start is off, the lift stop immediately

2.2 Normal run

Normal Run

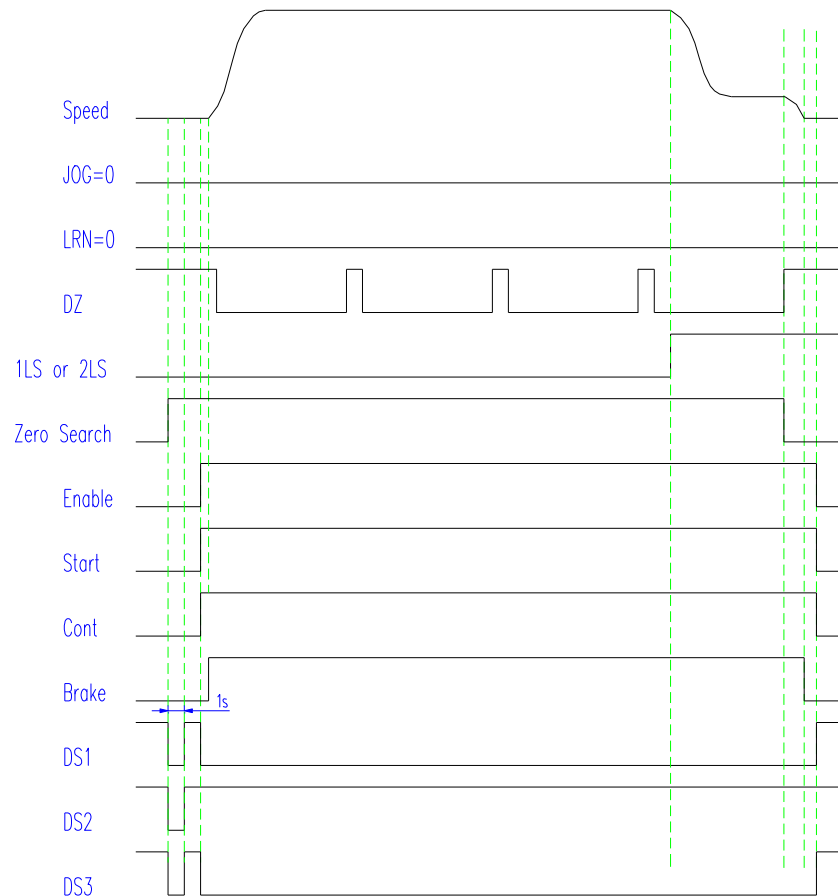


- AMCB2 should send the IP signal to LCBII before the lift reach the deceleration point. LCBII will count the floor number according to received IP signal during normal run, and if the next floor is stop floor, the LCBII will send the slow down (V codes) to AMCB2, AMCB2 will slow down followed according to curve and stop.
- When the run speed is less than the “door open speed threshold”, AMCB2 will sent the DS codes (DS1=1, DS2=1, DS3=0) to LCBII.

西子奥的斯电梯有限公司

2.3 Correct run

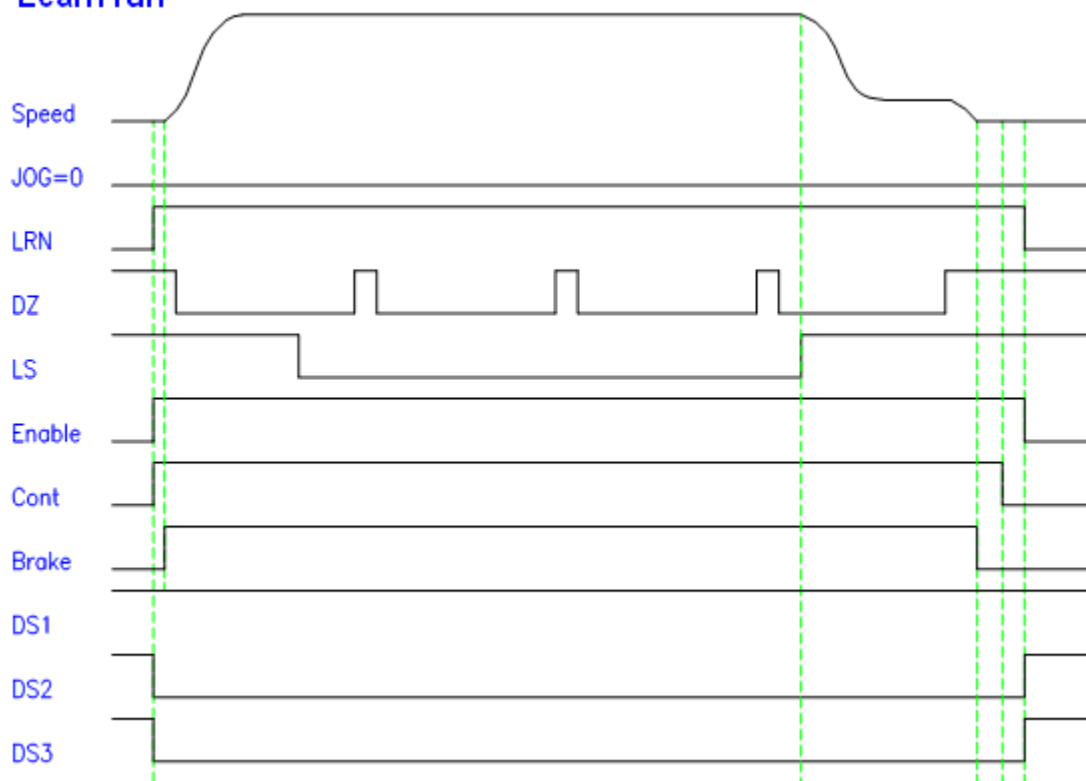
Zero Search Run (neither in bottom DZ nor in top DZ)



- When the AMCB2 or the LCBII lose the position, the lift enters the correct run mode. if AMCB2 lose the position, the lift run at half normal speed; and if only the LCBII lose the position, the lift run at normal speed.

2.4 Learning run

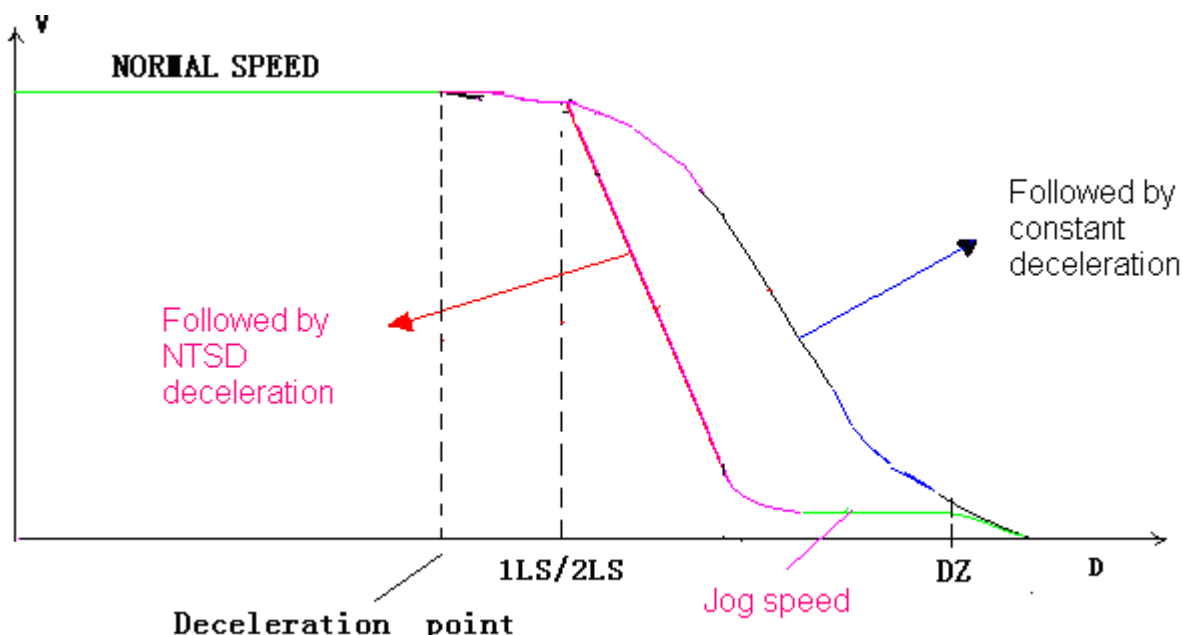
Learn run



- When the lift start to learn run, the car run to DZ of bottom floor with normal speed, then car run up with learning speed, and finished when the car get DZ of top floor.
- After finished the learning run, the lift return to normal mode automatically
- By learning run, AMCB2 get the floor position and 1LS/2LS position, you can find those in P menu.
- the speed of learning is same with inspection run
- AMCB2 do not sent the IP signal to LCBII in the course of learning run
- During the learning run, if let JOG to 1, the car quit the learning.

西子奥的斯电梯有限公司

2.5 NTSD protection run



- If the car hit the 1LS or 2LS at contract speed, the lift will be forced to perform an NTSD slowdown.

3 Parameters list introduce

3.1 System parameter F0

Number	Value	LCD Display	Description
F0 (Function)	F0 = 1	Enter password	Enter password, the initialize password: 1234
	F0 = 2	Change password	Change password
	F0 = 3		Set to 3, the terminal of Velocity Analog will output 10vdc
	F0 = 4	Shaft learning	Set to 4, lift will be enter learn mode
	F0 = 5	Reset parameter	Initialize the parameters

- The initialize password of AMCB2 is 1234; allowed to modify the parameters of F menu after affirming the password every time power on.

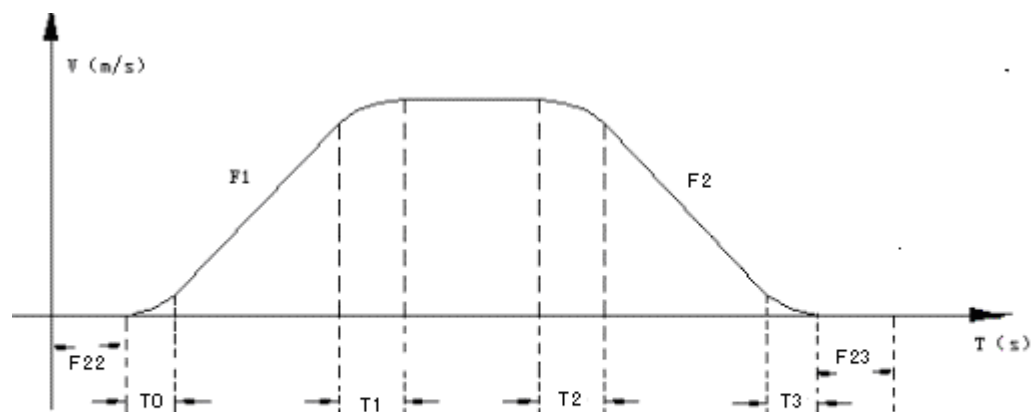
西子奥的斯电梯有限公司

- In order to ensure velocity display of AMCB2 and drive are the same, the gain and offset tune of velocity analog should be done before the car to normal run
- Before car to normal run, the hoistway learning run should be done; and before start to hoistway learning run, the blew items should be conformed:
the parameter of F24 should set to 0
the inspection switch is on normal position
and then set F0 to 4, the car start to hoistway learning run, after finished the hoistway learning, the F0 set to 1 automatically

3.2 Parameters of run curveF1~F6

Number	Display	Description	Range	Unit
F1	Acceleration	Sets the acceleration speed	200~1200	mm/s ²
F2	Deceleration	Sets the deceleration speed	200~1200	mm/s ²
F3	Acceleration T0	S-curve characteristic time T0~T3	500~1500	ms
F4	Acceleration T1		500~1300	ms
F5	Deceleration T2		500~1300	ms
F6	Deceleration T3		500~1500	ms

- The AMCB2 can set respectively the acceleration time and deceleration time; and the S-curve characteristic time also can be set respectively for beginning acceleration, ending acceleration, beginning deceleration and ending deceleration.
- the relation between F1~F6 is shown below:



Run cruve of lift

3.3 Related parameters about floor, speed and so on

Number	Display	Description	Range	Unit
F7	Floor number	the floor number	2~32	floor
F8	Rated speed	Set lift contract speed	200~2500	mm/s
F9	Encoder pulse	Set the number of encoder pulse	100~10000	PPR
F10	Motor rated RPM	Set the motor rated RPM	50~5000	RPM
F11	Inspection speed	Inspection speed	0~500	mm/s
F12	Open door	Set ADO speed	0~300	mm/s
F13	Stop speed	Stop speed	0~150	mm/s
F14	Releveling speed	Set the RLV speed	10~100	mm/s
F26	Setanalog speed	Normal speed gain	50~100	%
F27	Startup speed	Startup speed	0~30	mm/s
F28	Startup time	Startup time	0~1000	ms

- F7: the floor number, set according to the number of vane of DZ.

西子奥的斯电梯有限公司

- F8: contract speed ; F11: inspection and hoistway learn speed; F12: the speed of sending the LZ signal; F13: stop speed; F14: The speed of RLV.
- F9: the number of encoder pulses, when the pulses that the AMCB2 received is division output, the set F9 to encoder pulse/division ratio. E.g.:
the encoder pulses is 1024, set the division ratio(set by drive) to 4, you should set the F9 to 256(1024/4)
- F10: the motor RPM when the lift is running with contract speed.
- F26: the rated speed gain, the output speed of AMCB2 is F26XF8
- F27/F28: the startup speed and startup time, you can set the two parameters to improve the start performance(usually set F27 to 4mm/s, F28 to 300ms)

3.4 The parameters about adjustment of leveling position

Number	Display	Description	Range	Unit
F15	Upward adj	Upward leveling offset	0~200	mm
F16	Downward adj	Downward leveling offset	0~200	mm
F25	Level delay	Level delay offset	0~200	mm

- Before the adjustment of leveling, it should be sure that the leveling indicator plate of door zone has been mechanically adjusted before performing electrical leveling adjustment, and the car can run with high speed
- The car is half-loaded.
- Move downwards and record the error of leveling of each floor, adjust F16 Downward Adj. according to the average value of the error. (the value of F16 Downward Adj should be reduced if the elevator overshoots the switches, if not, it should be increased.)

西子奥的斯电梯有限公司

- Move upwards and record the error of leveling of each floor, adjust F15 Upward Adj. according to the average value of the error. (the value of F15 Downward Adj should be reduced if the elevator overshoots the switches, if not, it should be increased.)

3.5 The parameters of adjusting load weighting device

Number	Display	Description	Range	Unit
F17	Vpre-torque 100%	Output volt when 100% load	900~1000	10mV
F18	Vpre-torque 80%	Output volt when 80% load	700~900	10mV
F19	Vpre-torque 50%	Output volt when 50% load	350~750	10mV
F20	Vpre-torque 25%	Output volt when 25% load	150~400	10mV
F21	Vpre-torque 10%	Output volt when 10% load	50~150	10mV

- DRIVE=0(SIEI Inverter)

The AMCB2 can connect 5 load weighting switches most, when the relevant switches is activated, the AMCB2 will output relevant voltage by LWD analog output (terminal JP9-7 and JP9-8), the relation between the switches and output voltage as below:

Switch Output ovlt	ANSS (10%)	25L (25%)	50L (50%)	LNS (80%)	LWS (100%)
F21(0.5V—1.5V)	●	X	X	X	X
F20)1.5V—4V)	※	●	X	X	X
F19(3.5V—7.5V)	※	※	●	X	X
F18(7V—9V)	※	※	※	●	X
F17(9V-10V)	※	※	※	※	●

- Note: ●:the switch is on; X: the switch is off ; ※: Neglect the status of switch ;
- DIRVE=1(YAKAWA Inverter)

the AMCB2 is only permitted to connect 3 load weighting switched(25L,50L,80L).

西子奥的斯电梯有限公司

- F17~F21 used to adjust the output voltage when the relevant switch is on

3.6 Parameters about adjusting the brake delay:

Number	Display	Description	range	Unit
F22	Brake delay1	Brake delay time when the lift start	0~2000	ms
F23	brake delay2	Brake delay time when the lift stop	0~3000	ms
F32	Brake delay3		0~700	ms

4 Other function parameters

Number	Display	Description	range	Unit
F24	Run enable	Enable normal run	0: forbid normal run, inspection and learn run is permitted 1:normal run is permitted	
F29	Invert Type	Drive type	0: SIEI drive 1: vacant 2: YASKAWA	
F30	Disable LZ	Disable LZ	0: with LZ 1:without LZ	
F31	Brake SW type	Brake switch type	0: without brake switch 1: brake switch type is NO 2: brake switch type is NC	
F33	DDP time	Run timeout protection	1~45	s
F34	ZS Deceleration	Deceleration of correct run	100~900	mm/s ²

西子奥的斯电梯有限公司

5 Troubleshooting

there are only 6 faults that can be shown in displayer; the faults table is followed:

Display	Meaning	Probable cause	Corrective action
Inerter Fault	Inverter fault	Drive has not output signal of "DRIVE OK" The wiring of it is incorrect	Deal with the drive fault Fix the wiring
Encoder Fault	Encoder fault	The wiring of drive to AMCB2 is incorrect Noise interfere	Fix the wiring Check the signal noise
Level Fault	Level fault	The sensor is damaged The sequence of two sensors is reverse	Replace the sensor Change the wiring JP3-1 and JP3-2
CNT fbk Fault	Contactor feedback fault (DRIVE=2)	The contactor is damaged The wiring of it is incorrect	Replace the contactor Fix the wiring
BY fbk Fault	Brake feedback fault (DRIVE=2)	Brake relay is damaged The wiring of it is incorrect	Replace the relay Fix the wiring
DDP timeout:	DDP protection	The value is too small Machine is not run	Check the value Check the mechanical system